

2022-23 EC2R-July

ISM

A) 1) a) A = event of student passing exam
B = event of getting PPO

$$P(A) = \frac{3}{4}$$

$$P(B) = \frac{1}{8}$$

(independent events)

$$P(A \cap B) = \frac{3}{4} \times \frac{1}{8} = \frac{3}{32}$$

$$i) P(A \cap B) = \frac{3}{32}$$

$$ii) P(A') P(B) = \frac{1}{4} \times \frac{1}{8} = \frac{1}{32}$$

$$iii) P(A') P(B') = \frac{1}{4} \times \frac{7}{8} = \frac{7}{32}$$

b)

X	25	35	45	55	65	75	85
F(x)	10	20	25	35	5	5	0

$$n = 100$$

$$i) P(X > 55) = \frac{10}{100} = \frac{1}{10}$$

$$ii) P(X < 35) = \frac{10}{100} = \frac{1}{10}$$

$$iii) P(35 < X < 65) = \frac{25+35}{100} = \frac{3}{5}$$

~~B) 1) a)~~

A) 2) a)

$$P(A) = 0.3$$

$$P(B) = 0.5$$

$$P(A \cap B) = 0.15$$

$$P(A \cup B) = 0.3 + 0.5 - 0.15 = 0.65$$

$$i) P(A|A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P(A)}{P(A \cup B)} = \frac{0.3}{0.65} = \frac{6}{13} \approx 0.462$$

$$ii) P(A|A \cap B) = \frac{P(A \cap (A \cap B))}{P(A \cap B)} = \frac{P(A \cap B)}{P(A \cap B)} = 1$$

$$iii) P(A \cap B|A \cup B) = \frac{P((A \cap B) \cap (A \cup B))}{P(A \cup B)} = \frac{P(A \cap B)}{P(A \cup B)} = \frac{0.15}{0.65} = \frac{3}{13} \approx 0.231$$

2) b) A = delivery boy A

B = " " B

C = " " C

T = take more time than expected.

$$P(A) = 30\% = 0.3$$

$$P(T|A) = 5\% = 0.05$$

$$P(B) = 40\% = 0.4$$

$$P(T|B) = 10\% = 0.1$$

$$P(C) = 30\% = 0.3$$

$$P(T|C) = 3\% = 0.03$$

$$a) P(T) = P(A)P(T|A) + P(B)P(T|B) + P(C)P(T|C)$$

$$= \frac{8}{125} \approx 0.064$$

$$b) P(A|T) = \frac{P(A)P(T|A)}{P(T)} = \frac{0.3 \times 0.05}{0.064} = \frac{15}{64} \approx 0.234$$

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y \ x	x			
	0	1	2	3
0	0.15	0.3	0.05	0
1	0.05	0.15	2k	0.05
2	0	0.05	0.1	k

$$i) \sum_{x,y} P(x,y) = 1$$

$$\Rightarrow (0.15) \times 2 + (0.05) \times 4 + 0.1 + 0.3 + 3k = 1$$

$$\Rightarrow 3k = 0.1$$

$$\Rightarrow k = 0.0333$$

$$ii) \begin{array}{c|c|c|c|c} X & 0 & 1 & 2 & 3 \\ \hline P(X) & 0.2 & 0.5 & 0.216 & 0.0833 \end{array}$$

$$iii) \begin{array}{c|c|c|c} Y & 0 & 1 & 2 \\ \hline P(Y) & 0.5 & 0.316 & 0.1833 \end{array}$$

b)

$$\begin{array}{c|c|c|c|c} X & -2 & -1 & 1 & 2 \\ \hline P(X) & 0.2 & 0.4 & 0.25 & 0.15 \end{array}$$

i) $X \in R$, so, X being -ve doesn't invalidate the distribution. This is a valid distribution

$$\text{of } \sum_x P(x) = 1$$

$$ii) \mu = 0.2 \times (-2) + 0.4 \times (-1) + 0.25 + 0.15(2)$$

$$= \boxed{0.55} - 1/4$$

$$\sigma^2 = \sum_{x_i} (x_i - \mu)^2 P(x_i)$$

$$= 0.2(-2 - 0.25)^2 + 0.4(-1 - 0.25)^2 + 0.25(1 - 0.25)^2$$

$$+ 0.15(2 - 0.25)^2$$

$$= 0.2125$$

$$111) P(X > -1) = 0.25 + 0.15 = 0.4$$

$$B) 3) a) i) P(Y > X) = P(Y=1, X=0) + P(Y=2, X=0) + P(Y=2, X=1) \\ = 0.05 + 0 + 0.05 = \boxed{0.1}$$

$$ii) P(X=0, Y=0) = 0.15$$

$$P(X=0) = 0.2 \quad P(Y=0) = 0.5$$

$$P(X=0)P(Y=0) = 0.1 \neq P(X=0, Y=0)$$

$\therefore$ not independent.

$$3) 6) ii) E(X) = \mu = -1/4 = -0.25$$

$$E(X^2) = \sum x_i^2 P(x_i) =$$

$$= 0.2 \times (-2)^2 + 0.4 \times (-1)^2 + 0.25 + 0.15 \times (2)^2$$

$$= 0.8 + 0.4 + 0.25 + 0.6$$

$$= \boxed{2.05}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= 2.05 - (-0.25)^2$$

$$= \boxed{1.9875}$$

18/11/24

$$A) 9) a) n = 500 \\ p = 0.25$$

(normal approx for binomial)

$$\lambda = \mu = np = 125$$

$$\lambda = \sigma = \sqrt{np(1-p)} = 9.682$$

$$i) P(X > 290) = 1 - P(Z < \frac{290 - 125}{9.682})$$

$$= 1 - P(Z < 17.04)$$

$$= 1 - 1 = \boxed{0}$$

$$P(Z < 17.04) \approx 1$$

$$ii) P(X=250) = {}^{500}C_{250} \times (0.25)^{250} (0.75)^{250}$$

$$\approx 0$$

$$\begin{aligned} iii) P(120 < X < 180) &= P\left(Z < \frac{180-125}{9.682}\right) - P\left(Z < \frac{120-125}{9.682}\right) \\ &= P(Z < 5.68) - P(Z < -0.516) \\ &= 1 - 0.30153 \\ &= \boxed{0.69847} \end{aligned}$$

1) 4) b)

$$\begin{aligned} \mu &= 12 \\ \sigma &= 1.25 \end{aligned}$$

(normal distribution)

$$\begin{aligned} i) P(X > 15) &= 1 - P\left(Z < \frac{15-12}{1.25}\right) \\ &= 1 - P(Z < 2.4) \\ &= 1 - 0.9918 \\ &= \boxed{0.0082} \end{aligned}$$

$$\begin{aligned} ii) P(13 < X < 18) &= P\left(Z < \frac{18-12}{1.25}\right) - P\left(Z < \frac{13-12}{1.25}\right) \\ &= P(Z < 4.8) - P(Z < 0.8) \\ &= 1 - 0.7881 \\ &= \boxed{0.2119} \end{aligned}$$

$$iii) P(X=12) = 0$$